

Air Pollution Levels and Health Outcomes on Hospital Emergency Department Visits for HCA Healthcare Resource Allocation and Facility Preparation

Business Case

HCA Healthcare is the largest health system in the United States by number of hospitals, number of beds, and net patient revenue with 219 hospitals and more than 41,600 beds. The growing concern around air pollution and its impact on public health has become an urgent issue, especially as hospitals face increasing pressure to handle air pollution-related health problems. For HCA Healthcare, understanding the trend of hospital visits due to the effects of air pollution on hospital emergency room visits is vital for future demands on healthcare resources and preparation. Several studies have linked air pollution to hospitalizations for respiratory and cardiovascular issues, and as pollution levels rise, the frequency of hospital visits for conditions such as asthma, bronchitis, and heart disease may also increase.

The central business question guiding this study is: ***How can HCA Healthcare leverage predictive analytics to optimize resource allocation and facility preparedness across its 219 hospitals in response to air quality-driven visits?*** This question focuses on understanding how to ensure that HCA Healthcare can adequately prepare and respond to these cases. HCA Healthcare serves diverse geographic regions with varying air quality challenges and is responsible for managing a significant portion of the U.S. healthcare industry. Higher admission rates during high air pollution levels affects HCA hospitals' patient and revenue strategy, patient outcomes, direct costs (staffing optimization, supply chain management, equipment and technology), and operational efficiency.

The central analytics question we are asking is: ***What are the predictive relationships between environmental factors and health-related factors on hospital emergency room visits?*** This quantitative analysis aims to predict emergency room visits for pollution-related conditions, identify key air quality predictors, and provide actionable predictions that will enable HCA Healthcare to allocate resources effectively during periods of poor air quality. By addressing this question, the analysis will empower HCA Healthcare to proactively manage air pollution-related healthcare demands, ensuring operational resilience and improved patient care.

Dataset

This study utilizes a comprehensive dataset collected between April 2018 and March 2019 from Maharaj Nakorn Chiang Mai Hospital, a tertiary care and university hospital in Thailand. While this represents data from a single institution, the patterns and relationships identified can provide valuable insights for HCA Healthcare's operations, particularly in regions with similar air quality challenges.

The dataset encompasses both environmental and health outcome variables, providing a rich foundation for analysis. Environmental measurements include: fine particulate matter (PM_{2.5}) - particles less than 2.5 micrometers in diameter, coarse particulate matter (PM₁₀) - particles less than 10 micrometers in diameter, ozone (O₃) levels, nitrogen dioxide (NO₂) levels, and sulfur dioxide (SO₂) levels. These pollutants were measured daily and represent the most common air quality indicators used in public health research. The health outcome variables tracked include: daily emergency department (ED) visits, daily admission rates, cases categorized by specific conditions (acute respiratory disease (ARD), acute

coronary syndrome (ACS), acute heart failure (AHF), and stroke), total cases, and deaths. Please see *Appendix A* for a detailed description of each variable¹.

Descriptive Analytics

Initial analysis revealed several key patterns in hospital utilization and air quality relationships. On average, the hospital experienced 50 ED visits per day, with an admission rate of 9.2%, indicating that approximately 4–5 patients require inpatient care daily². This baseline understanding is crucial for resource planning and capacity management.

Among the primary health conditions studied, ARD showed the highest frequency, with an average of 5.4 cases per day, followed by stroke at 2.0 cases and ACS at 1.3 cases³. These conditions are critical from a business perspective, as they represent the most frequent and resource-intensive cases requiring hospital attention. Furthermore, the prevalence of ARD cases aligns with existing research that suggests respiratory conditions are particularly sensitive to air quality changes.

Correlation analysis⁴ revealed nuanced relationships regarding environmental factors: while individual pollutants showed relatively weak direct correlations with ED visits, their combined effects of multiple pollutants were more substantial. For instance, PM2.5 showed the strongest correlation with respiratory cases, and ozone levels were moderately associated with cardiovascular admissions. These findings suggest that the interactions among pollutants and their cumulative effects significantly influence hospital visit patterns. Seasonal patterns in ED visits were also evident, with higher rates corresponding to periods of poor air quality.

Pre-processing of the dataset included several steps before more robust analysis methods. For the most part, the dataset was clean. O2 had five missing values, and the calculated average (mean) was used to fill in the gaps. In later models, missing values were removed if necessary to maintain consistency and reliability, and variables were centered to reduce multicollinearity and improve model interpretability. Log transformations were applied to non-linear variables, such as the total case counts, to improve model fit. Lagged variables were generated to capture delayed effects of pollutants on health outcomes, reflecting the time-sensitive nature of environmental exposures.

Analytics Methods and Model Specifications

The analytical approach evolved through several stages of determining the best model, starting with a basic OLS Regression Model using raw variables⁵. While this initial model explained approximately 23.99% of the variance in ED visits (Multiple R-squared = 0.2399 and Adjusted R-square = 0.2206), diagnostic plots revealed that its performance was potentially constrained by some outliers, some concerning deviations at both tails, some heteroscedasticity and clustering patterns, and some residual non-normality⁶. However, the Variance Inflation Factor (VIF) test revealed that all values were < 10 and therefore no individual predictor exhibited extreme multicollinearity⁷. A Breusch-Pagan (BP) test

¹ Appendix A

² Appendix A

³ Appendix A

⁴ Appendix B

⁵ Appendix C

⁶ Appendix D

⁷ Appendix D

determined no heteroscedasticity (p-value = 0.1091), and a Durbin-Watson (DW) test indicated no serial correlation (D-W Statistic = 1.8025)⁸.

Three different modeling approaches were used to identify the best method for predicting daily case totals: a Time Series Model with lagged effects, a Log-Transformed Model, and an Interaction Model.

Because the dataset collected one year of data, model selection began with a Multivariate Time Series Model. This selection was to ensure that prediction over time included other predictors and control variables. Variables for both health conditions and environmental factors were lagged up to 3 days but ultimately did not improve the OLS Regression Model, decreased the R-squared values, and resulted in fewer significant predictors. A Stepwise Selection to simplify the model improved efficiency but still offered limited explanatory power⁹. The stepwise selection process identified seven key predictors: current-day effects of PM2.5, PM10, and O3 and lagged effects of PM2.5 (one-day), PM10 (two-day), O3 (three-day), and NO2 (three-day). The ANOVA test between the Multivariate Time Series Model and the Stepwise Model suggested the Stepwise Model achieved similar performance with fewer predictors, making it simpler and more interpretable¹⁰ but also highlighted the need for a more robust approach to capture complex relationships in the dataset.

The final analytical model framework employed a Log-Transformed Model¹¹ with interactions (Final Model), which substantially improved the model's explanatory power. Predictor variables were centered to reduce multicollinearity and improve interpretation of interaction effects, and the total case count was Log-transformed to address potential non-linearity. The Final Model emerged as the best modeling method because of its substantially higher explanatory power (Adjusted R-squared increased to 94.39%), capture of the complex relationships between pollutants and health outcomes, and satisfactory diagnostic results¹² (VIF scores < 10, BP Test p-value = 0.344, and D-W statistic = 1.949).

The visual inspection of the plots for the Final Model showed some curvilinear relationships despite the good diagnostic results. Attempts were made to correct the curvilinearity through a Polynomial Model¹³ and a Splines Model¹⁴. However, ANOVA Test results¹⁵ proved that neither model showed noteworthy improvement to the Final Model.

Cross-Validation testing¹⁶ was performed to validate the Final Model's predictive accuracy. The model's predictive performance was validated using bootstrapping with 25 samples. The results strongly support the model's predictive accuracy. The Root Mean Square Error (RMSE) = 0.0992, which represents the average prediction error in log units (i.e., an approximate 9.92% error in predicting total cases). This is a small error given the scale of the data. The cross-validated R-squared = 0.9374 (i.e., model explains approximately 93.74% of the variance in out-of-sample predictions). This is close to the Final Model's Adjusted R-squared of 0.9439, suggesting the model is not overfitting. The Mean Absolute Error (MAE) is 0.0671 and also measured in log units. It indicates that the prediction errors are approximately 6.71%, and because the MAE is lower than the RMSE, it shows that the predictions are generally consistent with

⁸ Appendix D

⁹ Appendix F

¹⁰ Appendix G

¹¹ Appendix H

¹² Appendix H

¹³ Appendix M

¹⁴ Appendix N

¹⁵ Appendix O

¹⁶ Appendix L

few extreme errors. The Cross-Validation results provide strong evidence that the model will be reliable for predicting hospital visits based on the air pollution and health outcome variables.

Results

Health conditions were the strongest predictors of total cases, with each condition showing substantial impact:

- ARD: 10.09% increase per unit (p-value < 0.001 ***)
- ACS: 9.93% increase per unit (p-value < 0.001 ***)
- AHF: 10.09% increase per unit (p-value < 0.001 ***)
- Stroke: 10.88% increase per unit (p-value < 0.001 ***)

The model identified significant negative interactions between pollutants and health conditions:

- PM2.5 × ARD: $\beta = -0.00016$ (p-value < 0.01 **)
- PM2.5 × AHF: $\beta = -0.00040$ (p-value < 0.001 ***)
- PM2.5 × Stroke: $\beta = -0.00031$ (p-value < 0.001 ***)
- O3 × ARD: $\beta = -0.00121$ (p-value < 0.001 ***)

These negative interactions suggest that when certain health conditions are already prevalent, the effect of air pollutants on total cases is less pronounced than expected. This could indicate that during periods of poor air quality, patients with these conditions may be taking preventive measures (such as staying indoors or seeking early medical attention) or that hospitals are implementing proactive interventions for high-risk patients.

From a business perspective, these insights are crucial for HCA Healthcare's resource planning. While higher pollution levels generally increase patient volume, the interaction effects suggest that the relationship is complex and varies by condition. Knowing this will help hospitals to better anticipate staffing needs during poor air quality days, prepare specific resources for expected increases in respiratory and cardiovascular cases, and implement targeted interventions for high-risk patients when air quality deteriorates.

Conclusions

The comprehensive analysis leads to several important conclusions about the relationship between air pollution and hospital resource demands, with significant implications for HCA Healthcare's operations and strategy.

While health conditions remain the primary drivers of ED visits, air pollution can be a significant moderating factor with nuanced effects. Different health conditions show varying sensitivities to air pollution levels, particularly evident in the interactions between PM2.5 and ARD, AHF, and stroke cases. This suggests that resource allocation strategies should be condition-specific, with air quality monitoring serving as a predictive tool. For example, when PM2.5 levels rise, hospitals should prepare for ARD and AHF cases, whereas elevated O3 levels particularly impact ARD cases.

The analysis reveals complex, non-linear relationships between pollutants and health outcomes, indicating that simple linear forecasting models may underestimate resource needs during extreme air quality events. To optimize operations, HCA Healthcare should implement real-time air quality monitoring systems across their regional hospital network, develop condition-specific resource allocation protocols triggered by air quality thresholds, and consider seasonal variations in staffing plans (as air quality concerns often

follow predictable patterns). These findings enable HCA Healthcare to move from reactive to proactive resource management, potentially improving both patient outcomes and operational efficiency.

Future analytical and strategic research should prioritize the key areas mentioned above when implementing the predictive power and practical use of the models. This study demonstrates the value of combining predictive analysis with practical business insights to improve healthcare decision-making and provides HCA Healthcare with a data-driven foundation for developing more sophisticated resource allocation strategies that account for both environmental factors and health outcomes.

Lessons Learned

Throughout this project, we encountered several challenges that shaped our modeling approach and analysis. Model selection proved to be the most challenging aspect during the process, marked by consistently low and/or decreasing R-squared values in our initial attempts before we centered the data. We explored nearly every transformation and correction approach, from Weighted Least Squares to Ridge and Lasso Regression. In one attempt, a Multivariate Time Series model resulted in a perfect multicollinearity issue with an Adjusted R-squared of 1. The lesson learned was that transforming the data itself can be more impactful than trying a different modeling approach.

In general, working with environmental and health variables presented its own set of challenges. We had to maintain focus on the requirements of the project itself rather than attempt to address broader climate science and public health questions. Finding the appropriate balance between variable transformations and interactions proved easier than we initially made it. At first, we considered between developing a logistic regression model versus creating a classification model that could serve as an alert system based on pollution levels like the Air Quality Index. We also experimented with various groups of variables, including seasonal categorizations (cool, hot, and rainy seasons) and a weighted pollution index that assigned different weights to various pollutants (PM2.5 at 60%, PM10 at 20%, and O3, NO2, and SO2 each at 5%).

Centering the data was the critical turning point for our modeling and analysis. We observed minimal improvements across different model iterations until then. As our Final Model, there were still some curvilinear relationships that we were unable to correct, but it showed us the complex nature of the relationships between the variables. Our full R code script is available in *Appendix P*.

The last challenge was that in our Business Case for Deliverables 1 and 2, we had planned to also analyze the relationship between the predictors and hospital admissions. Due to time limitations (and our sanity), we abandoned using it as a second response variable and will leave it to HCA Healthcare to handle in future analysis.

In the end, these challenges led us to our Final Model, which answered our analytics question by providing practical applicability of the dataset that can inform and aid HCA Healthcare with its strategic planning and decision-making.

Appendix A

Description of dataset

Variable	Range	Type	Unit	Description	R Class
Time Variables					
day	(1-365)	Discrete numeric	Day sequence number	Sequential day counter throughout study period	integer
date	m/d/y	Temporal	Calendar date	Daily date from April 2018 to March 2019	character
Disease-Specific Cases					
ard	(0-16)	Discrete numeric	Number of cases per day	Daily count of acute respiratory disease cases	integer
acs	(0-8)	Discrete numeric	Number of cases per day	Daily count of acute coronary syndrome cases	integer
ahf	(0-5)	Discrete numeric	Number of cases per day	Daily count of acute heart failure cases	integer
stroke	(0-8)	Discrete numeric	Number of cases per day	Daily count of stroke cases	integer
Hospital Metrics					
total_case	(2-23)	Discrete numeric	Number of cases per day	Total daily cases of the four specific conditions combined	integer
total_visit	(27-92)	Discrete numeric	Number of visits per day	Total daily Emergency Department visits	integer
admit	(0-13)	Discrete numeric	Number of admissions per day	Total daily hospital admissions	integer
death	(0-3)	Discrete numeric	Number of deaths per day	Total daily deaths	integer
Air Quality Metrics					
pm25	(39-282)	Continuous numeric	µg/m³	Daily PM2.5 concentration	integer
pm10	(9-176)	Continuous numeric	µg/m³	Daily PM10 concentration	integer
o3	(2-53)	Continuous numeric	ppb	Daily ozone concentration	integer
no2	(3-45)	Continuous numeric	ppb	Daily nitrogen dioxide concentration	integer
so2	(0-27)	Continuous numeric	ppb	Daily sulfur dioxide concentration	integer

Summary of dataset

```
> summary(data)
  day      date      ard      acs      ahf      stroke  total_case  total_visit  admit  death  pm25
Min.   : 1   Length:365   Min.   : 0.000   Min.   :0.000   Min.   :0   Min.   :0.000   Min.   : 2.000   Min.   :27.00   Min.   : 0.000   Min.   :0.0000   Min.   : 39.00
1st Qu.: 92   Class :character   1st Qu.: 3.000   1st Qu.:0.000   1st Qu.:0   1st Qu.:1.000   1st Qu.: 7.000   1st Qu.:44.00   1st Qu.: 3.000   1st Qu.:0.0000   1st Qu.: 63.00
Median :183   Mode  :character   Median : 5.000   Median :1.000   Median :1   Median :2.000   Median : 9.000   Median :49.00   Median : 4.000   Median :0.0000   Median : 75.00
Mean   :183                                     Mean   : 5.405   Mean   :1.288   Mean   :1   Mean   :2.005   Mean   : 9.699   Mean   :49.88   Mean   : 4.559   Mean   :0.4192   Mean   : 88.96
3rd Qu.:274                                     3rd Qu.: 7.000   3rd Qu.:2.000   3rd Qu.:2   3rd Qu.:3.000   3rd Qu.:11.000   3rd Qu.:55.00   3rd Qu.: 6.000   3rd Qu.:1.0000   3rd Qu.:102.00
Max.   :365                                     Max.   :16.000   Max.   :8.000   Max.   :5   Max.   :8.000   Max.   :23.000   Max.   :92.00   Max.   :13.000   Max.   :3.0000   Max.   :282.00

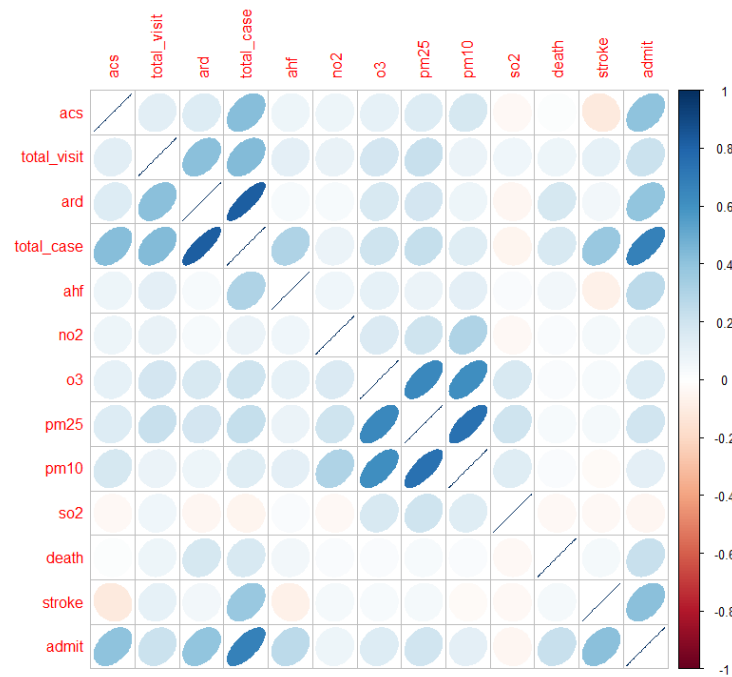
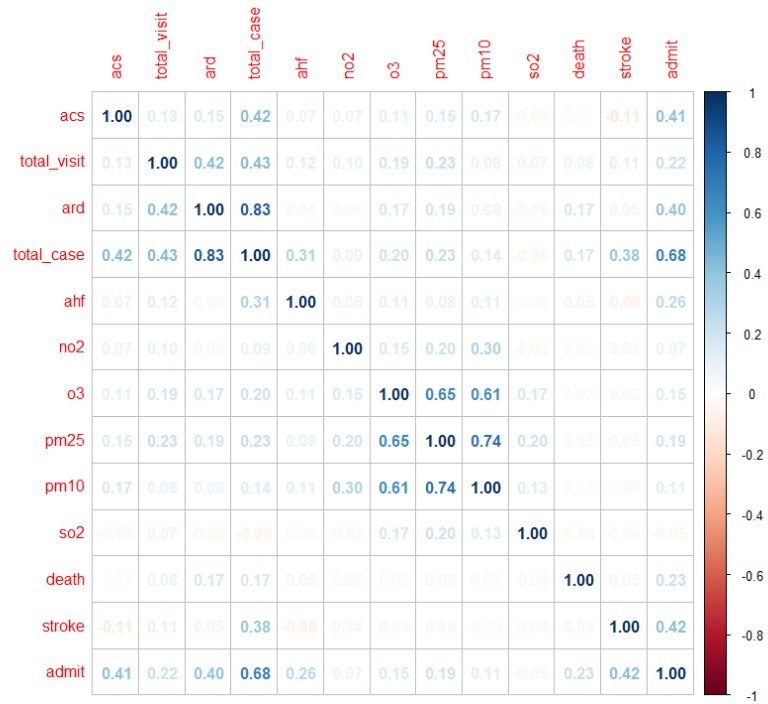
  pm10      o3      no2      so2
Min.   : 9.00   Min.   : 2.00   Min.   : 3.00   Min.   : 0.0000
1st Qu.:32.00   1st Qu.:15.00   1st Qu.: 6.00   1st Qu.: 0.0000
Median :39.00   Median :19.00   Median : 8.00   Median : 0.0000
Mean   :45.36   Mean   :19.93   Mean   :10.06   Mean   : 0.3753
3rd Qu.:53.00   3rd Qu.:24.00   3rd Qu.:12.00   3rd Qu.: 0.0000
Max.   :176.00   Max.   :53.00   Max.   :45.00   Max.   :27.0000
NA's   :5
```

Mean of hospital admissions

```
> mean(data$admit/data$total_visit)
[1] 0.0924034
```

Appendix B

Correlation Matrix



Appendix C

OLS Model summary with “total_visit” as response variable and pollution and health outcome variables as predictors

```
> OLSmodel <- lm(total_visit ~ ard + acs + ahf + stroke + pm25 + pm10 + o3 + no2 + so2, data = data)
> summary(OLSmodel)
```

Call:

```
lm(formula = total_visit ~ ard + acs + ahf + stroke + pm25 +
    pm10 + o3 + no2 + so2, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.206	-5.078	-0.256	4.664	38.850

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	38.43336	1.49160	25.766	< 2e-16	***
ard	1.13318	0.14751	7.682	1.54e-13	***
acs	0.48297	0.32949	1.466	0.14358	
ahf	0.78795	0.37704	2.090	0.03734	*
stroke	0.50046	0.27575	1.815	0.07038	.
pm25	0.04374	0.01585	2.759	0.00610	**
pm10	-0.08103	0.02961	-2.737	0.00652	**
o3	0.08234	0.07347	1.121	0.26320	
no2	0.09264	0.06460	1.434	0.15241	
so2	0.31635	0.24369	1.298	0.19508	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

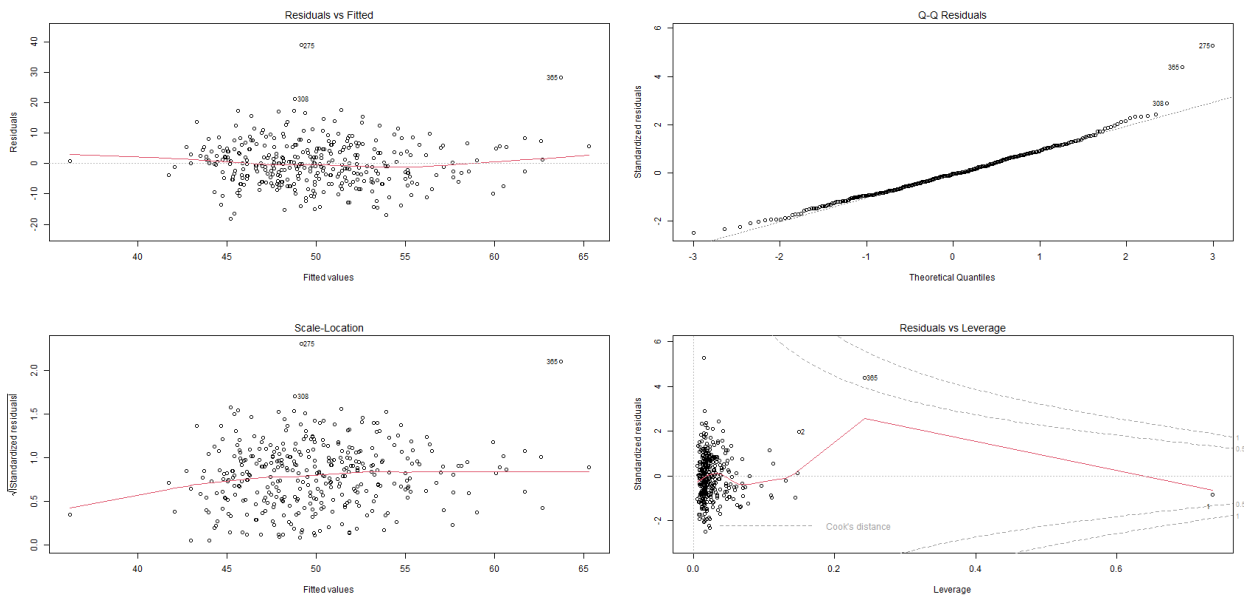
Residual standard error: 7.394 on 355 degrees of freedom

Multiple R-squared: 0.2399, Adjusted R-squared: 0.2206

F-statistic: 12.45 on 9 and 355 DF, p-value: < 2.2e-16

Appendix D

OLS Model Plots and Diagnostic Tests



```
> vif(OLSmodel)
      ard      acs      ahf      stroke      pm25      pm10      o3      no2      so2
1.080920 1.069871 1.024472 1.037529 2.707215 2.620607 1.863985 1.115462 1.066192
> bptest(OLSmodel)
```

studentized Breusch-Pagan test

```
data: OLSmodel
BP = 14.389, df = 9, p-value = 0.1091
```

```
> durbinWatsonTest(OLSmodel)
lag Autocorrelation D-W Statistic p-value
  1      0.07782565      1.802544    0.048
Alternative hypothesis: rho != 0
```

Appendix E

Lagged Model summary

```
Call:
lm(formula = total_case ~ pm25 + pm10 + o3 + no2 + so2 + pm25.L1 +
    pm25.L2 + pm25.L3 + pm10.L1 + pm10.L2 + pm10.L3 + o3.L1 +
    o3.L2 + o3.L3 + no2.L1 + no2.L2 + no2.L3 + so2.L1 + so2.L2 +
    so2.L3 + total_case.L1 + total_case.L2 + total_case.L3, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-7.427 -2.473 -0.268  1.921 11.985

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.837667   0.945710   7.230 3.24e-12 ***
pm25           0.015733   0.018732    0.840  0.4016
pm10          -0.021875   0.014360   -1.523  0.1286
o3             0.093512   0.042501    2.200  0.0285 *
no2            0.027089   0.068802    0.394  0.6940
so2           -0.332437   0.346864   -0.958  0.3385
pm25.L1        0.032191   0.020424    1.576  0.1159
pm25.L2        0.002659   0.020170    0.132  0.8952
pm25.L3       -0.001198   0.010964   -0.109  0.9131
pm10.L1        0.009352   0.026367    0.355  0.7231
pm10.L2       -0.052805   0.027343   -1.931  0.0543 .
pm10.L3       -0.001251   0.026771   -0.047  0.9627
o3.L1         -0.037802   0.046121   -0.820  0.4130
o3.L2         -0.024129   0.046209   -0.522  0.6019
o3.L3         -0.044296   0.043403   -1.021  0.3082
no2.L1        -0.031959   0.103532   -0.309  0.7577
no2.L2        -0.030385   0.104685   -0.290  0.7718
no2.L3         0.086156   0.072588    1.187  0.2361
so2.L1         0.046810   0.352909    0.133  0.8946
so2.L2         0.331128   0.320369    1.034  0.3021
so2.L3        -0.025140   0.161975   -0.155  0.8768
total_case.L1  0.056295   0.054681    1.030  0.3040
total_case.L2  0.067000   0.054337    1.233  0.2184
total_case.L3  0.003652   0.054748    0.067  0.9469
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.527 on 338 degrees of freedom
Multiple R-squared:  0.1098,    Adjusted R-squared:  0.04927
F-statistic: 1.813 on 23 and 338 DF,  p-value: 0.0134
```

Appendix F

Stepwise Selection Model summary

```
Call:
lm(formula = total_case ~ pm25 + pm10 + o3 + pm25.L1 + pm10.L2 +
    o3.L3 + no2.L3, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-7.0423 -2.6254 -0.3384  1.9098 12.3294

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.56429    0.64814   11.671  <2e-16 ***
pm25         0.01576    0.01115    1.414   0.1583
pm10        -0.02335    0.01368   -1.707   0.0887 .
o3          0.07963    0.03695    2.155   0.0318 *
pm25.L1     0.03757    0.01634    2.299   0.0221 *
pm10.L2    -0.05555    0.02472   -2.247   0.0252 *
o3.L3      -0.06137    0.03700   -1.659   0.0981 .
no2.L3      0.06206    0.03128    1.984   0.0480 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.484 on 354 degrees of freedom
Multiple R-squared:  0.09026,    Adjusted R-squared:  0.07227
F-statistic: 5.017 on 7 and 354 DF,  p-value: 0.00001943
```

Stepwise Selection Model Diagnostic Tests

```
> vif(Stepmodel)
      pm25      pm10      o3  pm25.L1  pm10.L2  o3.L3  no2.L3
5.926679 2.509969 2.089547 12.001096  7.289564  2.044208  1.166526
> bptest(Stepmodel)

studentized Breusch-Pagan test

data: Stepmodel
BP = 7.959, df = 7, p-value = 0.3362

> durbinWatsonTest(Stepmodel)
lag Autocorrelation D-W Statistic p-value
1      0.061367      1.873937    0.158
Alternative hypothesis: rho != 0
```

Appendix G

ANOVA Test for Lagged Model and Stepwise Model

```
> anova(Lagmodel, Stepmodel)
Analysis of Variance Table

Model 1: total_case ~ pm25 + pm10 + o3 + no2 + so2 + pm25.L1 + pm25.L2 +
  pm25.L3 + pm10.L1 + pm10.L2 + pm10.L3 + o3.L1 + o3.L2 + o3.L3 +
  no2.L1 + no2.L2 + no2.L3 + so2.L1 + so2.L2 + so2.L3 + total_case.L1 +
  total_case.L2 + total_case.L3
Model 2: total_case ~ pm25 + pm10 + o3 + pm25.L1 + pm10.L2 + o3.L3 + no2.L3
  Res.Df    RSS   Df Sum of Sq    F Pr(>F)
1      338 4205.6
2      354 4298.1 -16    -92.511 0.4647 0.9622
> |
```

Appendix H

Log-Transformed Model summary and Diagnostic Tests

```
Call:
lm(formula = log_total_case ~ pm25 + pm10 + o3 + o3.L2 + log_total_case.L1,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.40484 -0.22460  0.02072  0.23478  0.87784

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.8836462   0.1212645   15.533 < 2e-16 ***
pm25           0.0029366   0.0008621    3.406 0.000734 ***
pm10          -0.0022434   0.0014141   -1.586 0.113536
o3            0.0075596   0.0037905    1.994 0.046880 *
o3.L2         -0.0060110   0.0038204   -1.573 0.116524
log_total_case.L1 0.0585815  0.0522602    1.121 0.263063
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3663 on 355 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.07954,    Adjusted R-squared:  0.06658
F-statistic: 6.136 on 5 and 355 DF,  p-value: 0.0000181

> vif(Logmodel)
              pm25              pm10              o3              o3.L2 log_total_case.L1
              3.204901              2.426868              1.989570              1.998157              1.054185
> bptest(Logmodel)

              studentized Breusch-Pagan test

data:  Logmodel
BP = 3.0353, df = 5, p-value = 0.6945

> durbinWatsonTest(Logmodel)
lag Autocorrelation D-W Statistic p-value
 1      -0.01001107      2.016729    0.964
Alternative hypothesis: rho != 0
```

Appendix I

Log-Transformed Model with Interactions summary and Diagnostic Tests

```
Call:
lm(formula = log_total_case ~ pm25_c + o3_c + pm10_c + no2_c +
    so2_c + ard_c + acs_c + ahf_c + stroke_c + pm25_c:ard_c +
    pm25_c:acs_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
    o3_c:acs_c + o3_c:ahf_c + o3_c:stroke_c + log_total_case.L1,
    data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.58286 -0.02953  0.03088  0.05702  0.09779

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.25269403  0.02932561  76.817 < 2e-16 ***
pm25_c       0.00036427  0.00019688   1.850 0.065148 .
o3_c        -0.00071023  0.00092302  -0.769 0.442148
pm10_c       0.00013196  0.00036855   0.358 0.720521
no2_c       -0.00020738  0.00079212  -0.262 0.793628
so2_c       0.00348668  0.00831220   0.419 0.675139
ard_c       0.10147307  0.00184419  55.023 < 2e-16 ***
acs_c       0.09998768  0.00406615  24.590 < 2e-16 ***
ahf_c       0.10132641  0.00466891  21.702 < 2e-16 ***
stroke_c     0.10879175  0.00338112  32.176 < 2e-16 ***
log_total_case.L1 -0.01796624  0.01309695  -1.372 0.171029
pm25_c:ard_c -0.00014822  0.00006860  -2.161 0.031420 *
pm25_c:acs_c -0.00007418  0.00013864  -0.535 0.592940
pm25_c:ahf_c -0.00034458  0.00014795  -2.329 0.020436 *
pm25_c:stroke_c -0.00034095  0.00011603  -2.938 0.003523 **
o3_c:ard_c   -0.00121136  0.00036460  -3.322 0.000989 ***
o3_c:acs_c   -0.00088271  0.00075405  -1.171 0.242564
o3_c:ahf_c   -0.00048126  0.00077668  -0.620 0.535907
o3_c:stroke_c 0.00004489  0.00066873   0.067 0.946516
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08969 on 342 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.9468,    Adjusted R-squared:  0.944
F-statistic: 338.3 on 18 and 342 DF, p-value: < 2.2e-16

> vif(Interactmodel)
there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif
      pm25_c      o3_c      pm10_c      no2_c      so2_c      ard_c      acs_c      ahf_c      stroke_c log_total_case.L1
1 2.787112    1.967226    2.748703    1.137154    1.145712    1.139447    1.091088    1.040430    1.054449    1.104042
2 2.228316    1.964619    1.795292    2.116535    2.041100    1.768355    1.655241    1.983059
```

```
> bptest(Interactmodel)

studentized Breusch-Pagan test

data: Interactmodel
BP = 16.79, df = 18, p-value = 0.5376

> durbinWatsonTest(Interactmodel)
lag Autocorrelation D-W Statistic p-value
1      0.03860428      1.921373      0.388
Alternative hypothesis: rho != 0
```

Appendix J

Simplified Final Model summary and Diagnostic Tests

```
Call:
lm(formula = log_total_case ~ pm25_c + o3_c + pm10_c + no2_c +
    so2_c + ard_c + acs_c + ahf_c + stroke_c + pm25_c:ard_c +
    pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c + log_total_case.L1,
    data = data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.60493 -0.02857  0.02733  0.05434  0.11990
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.25056597  0.02929135   76.834 < 2e-16 ***
pm25_c        0.00035111  0.00019647    1.787 0.074798 .
o3_c         -0.00062660  0.00091670   -0.684 0.494725
pm10_c        0.00009012  0.00036728    0.245 0.806311
no2_c        -0.00013774  0.00079111   -0.174 0.861880
so2_c         0.00445839  0.00829669    0.537 0.591358
ard_c         0.10143608  0.00184141   55.086 < 2e-16 ***
acs_c         0.09932985  0.00405151   24.517 < 2e-16 ***
ahf_c         0.10095160  0.00466409   21.644 < 2e-16 ***
stroke_c      0.10884061  0.00336797   32.316 < 2e-16 ***
log_total_case.L1 -0.01758226  0.01308719   -1.343 0.180000
pm25_c:ard_c  -0.00016469  0.00006308   -2.611 0.009428 **
pm25_c:ahf_c  -0.00040309  0.00011655   -3.458 0.000611 ***
pm25_c:stroke_c -0.00031829  0.00008323   -3.824 0.000155 ***
o3_c:ard_c    -0.00121841  0.00035323   -3.449 0.000631 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '.' 0.05 ' ' 1
```

```
Residual standard error: 0.08977 on 346 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.9461,    Adjusted R-squared:  0.9439
F-statistic: 433.8 on 14 and 346 DF,  p-value: < 2.2e-16
```

```
> vif(Finalmodel)
there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif
      pm25_c      o3_c      pm10_c      no2_c      so2_c      ard_c      acs_c      ahf_c      stroke_c log_total_case.L1
pm25_c      2.770649      1.936976      2.724988      1.132246      1.139419      1.133999      1.081324      1.036443      1.044411      1.100446
pm25_c:ard_c      1.880845      1.112242      1.086990      1.912393
```

```
> bptest(Finalmodel)

studentized Breusch-Pagan test

data: Finalmodel
BP = 15.511, df = 14, p-value = 0.3441

> durbinWatsonTest(Finalmodel)
lag Autocorrelation D-W Statistic p-value
 1      0.02472321      1.949329      0.492
Alternative hypothesis: rho != 0
```

Appendix K

ANOVA Test for Log-Transformed Model with Interactions and Simplified Final Model

```
> anova(Interactmodel, Finalmodel)
Analysis of Variance Table

Model 1: log_total_case ~ pm25_c + o3_c + pm10_c + no2_c + so2_c + ard_c +
  acs_c + ahf_c + stroke_c + pm25_c:ard_c + pm25_c:acs_c +
  pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c + o3_c:acs_c +
  o3_c:ahf_c + o3_c:stroke_c + log_total_case.L1
Model 2: log_total_case ~ pm25_c + o3_c + pm10_c + no2_c + so2_c + ard_c +
  acs_c + ahf_c + stroke_c + pm25_c:ard_c + pm25_c:ahf_c +
  pm25_c:stroke_c + o3_c:ard_c + log_total_case.L1
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     342  2.7512
2     346  2.7883 -4  -0.037115  1.1534 0.3313
```

Appendix L

Cross Validation Testing

```
Call:
lm(formula = .outcome ~ ., data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-0.60493 -0.02857  0.02733  0.05434  0.11990

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.25056597  0.02929135   76.834 < 2e-16 ***
pm25_c          0.00035111  0.00019647    1.787 0.074798 .
o3_c           -0.00062660  0.00091670   -0.684 0.494725
pm10_c          0.00009012  0.00036728    0.245 0.806311
no2_c           -0.00013774  0.00079111   -0.174 0.861880
so2_c           0.00445839  0.00829669    0.537 0.591358
ard_c           0.10143608  0.00184141   55.086 < 2e-16 ***
acs_c           0.09932985  0.00405151   24.517 < 2e-16 ***
ahf_c           0.10095160  0.00466409   21.644 < 2e-16 ***
stroke_c        0.10884061  0.00336797   32.316 < 2e-16 ***
log_total_case.L1 -0.01758226  0.01308719   -1.343 0.180000
`pm25_c:ard_c`   -0.00016469  0.00006308   -2.611 0.009428 **
`pm25_c:ahf_c`  -0.00040309  0.00011655   -3.458 0.000611 ***
`pm25_c:stroke_c` -0.00031829  0.00008323   -3.824 0.000155 ***
`o3_c:ard_c`     -0.00121841  0.00035323   -3.449 0.000631 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08977 on 346 degrees of freedom
Multiple R-squared:  0.9461,    Adjusted R-squared:  0.9439
F-statistic: 433.8 on 14 and 346 DF,  p-value: < 2.2e-16

> CVmodel
Linear Regression

361 samples
10 predictor

No pre-processing
Resampling: Bootstrapped (25 reps)
Summary of sample sizes: 361, 361, 361, 361, 361, 361, ...
Resampling results:

      RMSE      Rsquared    MAE
0.09928906  0.9374639  0.06716341

Tuning parameter 'intercept' was held constant at a value of TRUE
```

Appendix M

Polynomial Model summary and Plots

```
Call:
lm(formula = log_total_case ~ pm25_c + I(pm25_c^2) + o3_c + I(o3_c^2)
    pm10_c + I(pm10_c^2) + no2_c + so2_c + ard_c + acs_c + ahf_c +
    stroke_c + pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c +
    o3_c:ard_c + log_total_case.L1, data = data)
```

Residuals:

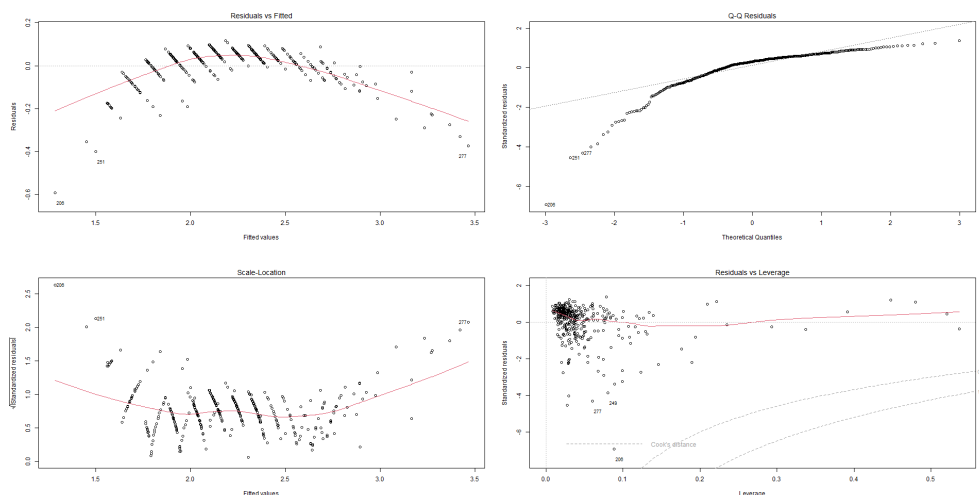
Min	1Q	Median	3Q	Max
-0.59159	-0.02948	0.02752	0.05216	0.11700

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.252499106	0.030723988	73.314	< 2e-16 ***
pm25_c	0.000019964	0.000284507	0.070	0.944099
I(pm25_c^2)	0.000003389	0.000002326	1.457	0.146087
o3_c	-0.000900207	0.000998213	-0.902	0.367786
I(o3_c^2)	0.000030601	0.000058753	0.521	0.602813
pm10_c	0.001008580	0.000536731	1.879	0.061076 .
I(pm10_c^2)	-0.000013961	0.000005973	-2.337	0.020006 *
no2_c	-0.000274781	0.000790236	-0.348	0.728264
so2_c	0.004900451	0.008284242	0.592	0.554549
ard_c	0.101581932	0.001859393	54.632	< 2e-16 ***
acs_c	0.099413995	0.004073302	24.406	< 2e-16 ***
ahf_c	0.100173332	0.004660192	21.496	< 2e-16 ***
stroke_c	0.109094583	0.003363537	32.434	< 2e-16 ***
log_total_case.L1	-0.018521645	0.013288231	-1.394	0.164269
pm25_c:ard_c	-0.000173363	0.000063321	-2.738	0.006507 **
pm25_c:ahf_c	-0.000454513	0.000119599	-3.800	0.000171 ***
pm25_c:stroke_c	-0.000341859	0.000083702	-4.084	0.0000551 ***
o3_c:ard_c	-0.001273512	0.000354382	-3.594	0.000374 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08943 on 343 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared: 0.947, Adjusted R-squared: 0.9444
F-statistic: 360.4 on 17 and 343 DF, p-value: < 2.2e-16



Appendix N

Splines Model and Plots

```
Call:
lm(formula = log_total_case ~ ns(pm25_c, df = 3) + ns(o3_c, df = 3) +
    ns(pm10_c, df = 3) + no2_c + so2_c + ard_c + acs_c + ahf_c +
    stroke_c + pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c +
    o3_c:ard_c + log_total_case.L1, data = data)
```

Residuals:

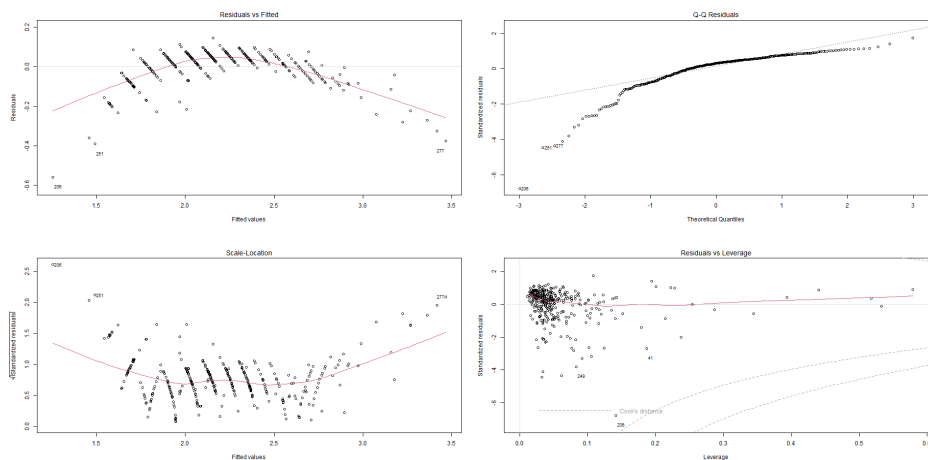
Min	1Q	Median	3Q	Max
-0.56033	-0.02835	0.02702	0.05207	0.14684

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.18160667	0.06565864	33.226	< 2e-16 ***
ns(pm25_c, df = 3)1	0.00998535	0.03908287	0.255	0.798498
ns(pm25_c, df = 3)2	0.02129439	0.06147927	0.346	0.729281
ns(pm25_c, df = 3)3	0.12148345	0.06431115	1.889	0.059743 .
ns(o3_c, df = 3)1	-0.04700486	0.03219530	-1.460	0.145216
ns(o3_c, df = 3)2	0.11565646	0.09169389	1.261	0.208054
ns(o3_c, df = 3)3	0.06480791	0.06551472	0.989	0.323263
ns(pm10_c, df = 3)1	0.08584783	0.04617130	1.859	0.063844 .
ns(pm10_c, df = 3)2	0.05955359	0.11598435	0.513	0.607961
ns(pm10_c, df = 3)3	-0.06959972	0.07052389	-0.987	0.324396
no2_c	-0.00028435	0.00080984	-0.351	0.725717
so2_c	0.00577019	0.00834209	0.692	0.489600
ard_c	0.10138196	0.00186403	54.389	< 2e-16 ***
acs_c	0.10007650	0.00407534	24.557	< 2e-16 ***
ahf_c	0.10081600	0.00465091	21.677	< 2e-16 ***
stroke_c	0.10947465	0.00336025	32.579	< 2e-16 ***
log_total_case.L1	-0.01901670	0.01331441	-1.428	0.154129
ard_c:pm25_c	-0.00019111	0.00006431	-2.972	0.003174 **
ahf_c:pm25_c	-0.00042372	0.00011890	-3.564	0.000418 ***
stroke_c:pm25_c	-0.00034605	0.00008448	-4.096	0.0000526 ***
ard_c:o3_c	-0.00108395	0.00036443	-2.974	0.003146 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08915 on 340 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared: 0.9478, Adjusted R-squared: 0.9447
F-statistic: 308.5 on 20 and 340 DF, p-value: < 2.2e-16



Appendix O

ANOVA Test for Polynomial Model and Splines Model

```
> anova(Finalmodel, Polymodel, Splinesmodel)
Analysis of Variance Table

Model 1: log_total_case ~ pm25_c + o3_c + pm10_c + no2_c + so2_c + ard_c +
  acs_c + ahf_c + stroke_c + pm25_c:ard_c + pm25_c:ahf_c +
  pm25_c:stroke_c + o3_c:ard_c + log_total_case.L1
Model 2: log_total_case ~ pm25_c + I(pm25_c^2) + o3_c + I(o3_c^2) + pm10_c +
  I(pm10_c^2) + no2_c + so2_c + ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1
Model 3: log_total_case ~ ns(pm25_c, df = 3) + ns(o3_c, df = 3) + ns(pm10_c,
  df = 3) + no2_c + so2_c + ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     346 2.7883
2     343 2.7431  3  0.045218 1.8963 0.1299
3     340 2.7024  3  0.040681 1.7060 0.1655
```

Appendix P

R code script

Data analysis

```
data <- read.csv("project.total.csv")
summary(data)
mean(data$admit/data$total_visit)
```

```
colSums(is.na(data))
data$o3[is.na(data$o3)] <- mean(data$o3, na.rm = TRUE)
```

```
corr.plot <- cor(data[, c("ard", "acs", "ahf", "stroke", "total_case", "total_visit", "admit", "death", "pm25",
"pm10", "o3", "no2", "so2")])
corrplot(corr.plot, method = "number", order = "hclust")
corrplot(corr.plot, method = "ellipse", order = "hclust")
```

#OLS Model

```
OLSmodel <- lm(total_visit ~ ard + acs + ahf + stroke + pm25 + pm10 + o3 + no2 + so2, data = data)
summary(OLSmodel)
par(mfrow = c(2, 2))
plot(OLSmodel)
vif(OLSmodel)
bptest(OLSmodel)
durbinWatsonTest(OLSmodel)
```

Lagged model

```
healthvars <- c("ard", "acs", "ahf", "stroke", "total_case")
for(var in healthvars) {
  for(lag in 1:3) {
    data <- slide(data, Var=var, NewVar=paste0(var, ".L", lag), slideBy=-lag)
  }
}
envvars <- c("pm25", "pm10", "o3", "no2", "so2")
for(var in envvars) {
  for(lag in 1:3) {
    data <- slide(data, Var=var, NewVar=paste0(var, ".L", lag), slideBy=-lag)
  }
}

colSums(is.na(data))
```

```

data <- na.omit(data)

Lagmodel <- lm(total_case ~
  pm25 + pm10 + o3 + no2 + so2 +
  pm25.L1 + pm25.L2 + pm25.L3 +
  pm10.L1 + pm10.L2 + pm10.L3 +
  o3.L1 + o3.L2 + o3.L3 +
  no2.L1 + no2.L2 + no2.L3 +
  so2.L1 + so2.L2 + so2.L3 +
  total_case.L1 + total_case.L2 + total_case.L3,
  data=data)
summary(Lagmodel)

# Stepwise selection

Stepmodel <- step(Lagmodel, direction="both")
summary(Stepmodel)
vif(Stepmodel)
bptest(Stepmodel)
durbinWatsonTest(Stepmodel)
par(mfrow=c(2,2))
plot(Stepmodel)

anova(Lagmodel, Stepmodel)

# Log transformation model

data$log_total_case <- log(data$total_case)
data <- slide(data, Var="log_total_case", NewVar="log_total_case.L1", slideBy = -1)
Logmodel <- lm(log_total_case ~ pm25 + pm10 + o3 + o3.L2 + log_total_case.L1,
  data=data)
summary(Logmodel)
vif(Logmodel)
bptest(Logmodel)
durbinWatsonTest(Logmodel)
par(mfrow=c(2,2))
plot(Logmodel)

# Center variables

data$pm25_c <- scale(data$pm25, center=TRUE, scale=FALSE)
data$o3_c <- scale(data$o3, center=TRUE, scale=FALSE)
data$pm10_c <- scale(data$pm10, center=TRUE, scale=FALSE)
data$no2_c <- scale(data$no2, center=TRUE, scale=FALSE)

```

```

data$so2_c <- scale(data$so2, center=TRUE, scale=FALSE)
data$ard_c <- scale(data$ard, center=TRUE, scale=FALSE)
data$acs_c <- scale(data$acs, center=TRUE, scale=FALSE)
data$ahf_c <- scale(data$ahf, center=TRUE, scale=FALSE)
data$stroke_c <- scale(data$stroke, center=TRUE, scale=FALSE)

# Interactions model

Interactmodel <- lm(log_total_case ~
  pm25_c + o3_c + pm10_c + no2_c + so2_c +
  ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:acs_c + pm25_c:ahf_c + pm25_c:stroke_c +
  o3_c:ard_c + o3_c:acs_c + o3_c:ahf_c + o3_c:stroke_c +
  log_total_case.L1,
  data=data)
summary(Interactmodel)
vif(Interactmodel)
bptest(Interactmodel)
durbinWatsonTest(Interactmodel)
par(mfrow=c(2,2))
plot(Interactmodel)

# Simplified Interactions model

Finalmodel <- lm(log_total_case ~
  pm25_c + o3_c + pm10_c + no2_c + so2_c +
  ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1,
  data = data)

summary(Finalmodel)
vif(Finalmodel)
bptest(Finalmodel)
durbinWatsonTest(Finalmodel)
par(mfrow=c(2,2))
plot(Finalmodel)

anova(Interactmodel, Finalmodel)

# Cross-validation test statistics

library(caret)
set.seed(1)

```

```

data <- na.omit(data)
CVmodel <- train(log_total_case ~ pm25_c + o3_c + pm10_c + no2_c + so2_c +
  ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1,
  data = data,
  method = "lm")
summary(CVmodel)
CVmodel

```

Trying to correct curvilinear relationships

```

Polymodel <- lm(log_total_case ~
  pm25_c + I(pm25_c^2) +
  o3_c + I(o3_c^2) +
  pm10_c + I(pm10_c^2) +
  no2_c + so2_c +
  ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1,
  data = data)
summary(Polymodel)
vif(Polymodel)
bptest(Polymodel)
durbinWatsonTest(Polymodel)
par(mfrow=c(2,2))
plot(Polymodel)

```

```

Splinesmodel <- lm(log_total_case ~
  ns(pm25_c, df=3) +
  ns(o3_c, df=3) +
  ns(pm10_c, df=3) +
  no2_c + so2_c +
  ard_c + acs_c + ahf_c + stroke_c +
  pm25_c:ard_c + pm25_c:ahf_c + pm25_c:stroke_c + o3_c:ard_c +
  log_total_case.L1,
  data = data)
summary(Splinesmodel)
vif(Splinesmodel)
bptest(Splinesmodel)
durbinWatsonTest(Splinesmodel)
par(mfrow=c(2,2))
plot(Splinesmodel)
anova(Finalmodel, Polymodel, Splinesmodel)

```